Ángel **Longueira Romero**

PhD - Industrial Cybersecurity - Embedded Systems and OT Security

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Industrial cybersecurity researcher and embedded systems security specialist with a PhD in Industrial Cybersecurity and experience in the **design, evaluation and deployment** of secure architectures for industrial and critical infrastructure systems.

Specialised in **vulnerability analysis** of embedded devices, cybersecurity **evaluation** methodologies, **PKI** infrastructures for industrial environments and **hardware-rooted security technologies** such as Trusted Execution Environments.

Experienced in applying **industrial cybersecurity standards and frameworks** including IEC 62443, NIS2, CRA and Common Criteria to real industrial systems.

Currently working at Tecnalia, supporting industrial partners and European R&D projects focused on secure architectures for smart grids, industrial devices and distributed infrastructures.

Technical Skills

Cybersecurity Standards — Application and interpretation of cybersecurity standards and regulatory frameworks for industrial and critical infrastructure systems, including IEC 62443, NIS2 and the Cyber Resilience Act (CRA).

Public Key Infrastructure (PKI) — Hands-on experience in the design and deployment for industrial environments, focusing on device identity. Implemented automated certificate lifecycle management using protocols such as SCEP, EST and OCSP, including integration with HSM. Familiar with large-scale device provisioning challenges and secure key management in embedded contexts.

Embedded Systems (OT) — Microcontroller-based platforms, embedded Linux environments and hardware-assisted security mechanisms. Low-level components such as firmware, secure boot processes and hardware security modules (TPM, TrustZone). Familiar with development and integration workflows in constrained environments, including Yocto-based systems and custom hardware platforms.

Smart Grids — Distributed architectures, device heterogeneity and real-time constraints. Understanding of communication models and protocols used in energy systems, as well as the security implications of integrating legacy and modern infrastructures.

Cybersecurity Evaluation — Experience in vulnerability assessment, threat modeling and security evaluation, incorporating metric-based approaches to assess security posture.

Programming — C, C++ and Python with experience in both high-level and low-level development contexts. Capable of working close to hardware in embedded environments. Comfortable working across Linux-based development environments.

Professional Experience

February 2023 until now — **Industrial Cybersecurity Researcher, TECNALIA Research and Innovation, Derio - Spain**

- Coordination of the hardware cybersecurity group.
- Preparation of funding proposals for European and Basque Government projects, including partner coordination and work package management.
- PKI design and deployment in AWS Cloud with HSM. Development and execution of the key ceremony procedure.
- Risk assessment and vulnerability analysis.
- Development of operating systems based on YOCTO.
- Trusted Applications development for SGX (Intel) and TrustZone (ARM) TEEs.
- Teaching Embedded Systems and Smart Grid Cybersecurity Workshop.

PKI · EJBCA · TEE · SGX · TrustZone · Cryptography · Embedded Systems · Smart Grids · IEC 62443 · CRA · NIS2 · IEC 61850 · HSM · TPM

May 2022 – February 2023
Industrial Cybersecurity Researcher, IKERLAN, Arrasate - Spain
- Preparation of funding proposals for Basque Government projects.
- Gap analysis for Cybersecurity Management Lifecycle based on IEC 62443.
- PKI design and deployment on premise with HSM. Development and execution of the key ceremony procedure.
- Secure architecture design for ROS2 industrial environments.

Gap Analysis | PKI | HSM | Cryptography | ROS2 | IEC 62443 | Cybersecurity Metrics | EJBCA | SCEP | EST | OCSP

March 2019 – May 2022
PhD Candidate, IKERLAN, Arrasate - Spain
- Cybersecurity evaluation methodologies for Embedded Systems.
- Cybersecurity metrics and metric aggregation.
- Dependency modeling based on Directed Graphs and Bayesian Networks.

Common Criteria | MAGERIT | Directed Graphs | Bayesian Networks | Metric Aggregation | Embedded Systems | IEC 62443 | NIST SP-800

September 2018 – March 2019
Industrial Cybersecurity Intern, IKERLAN, Arrasate - Spain
- Design of a laboratory-grade PKI.
- Deployment of a laboratory-grade PKI.
- Deployment and configuration of HSMs.
- Deployment and configuration of EJBCA (certificate lifecycle management software).

PKI | EJBCA | HSM | SCEP | EST | OCSP | Cryptography | Embedded Systems

July 2017 – June 2018
Amplification Equipment Repair Apprentice, Ortodoxo Amplification, Gijón - Spain
- Repair and restoration of vintage amplification equipment (valve amplifiers).
- Development of ad hoc tools to support the repair process.

Power Electronics | Valve Amplifiers

June 2016 – June 2017
Mentoring and Tutoring, Self-employed, Gijón - Spain
- Private lessons for high school students.
- Python Programming.
- Physics, mathematics and technology.

Teaching | Python | Mathematics | Physics

Education

2019 - 2022 PhD in Industrial Cybersecurity, Faculty of Engineering, Mondragon University:
- Cybersecurity Evaluation Standards.
- Vulnerability analysis.
- Cybersecurity metrics.
- Cum laude.

2017 - 2019 Master's degree in Automation Engineering and Industrial Informatics, Gijón Polytechnic School of Engineering:
- Optical inspection techniques for industrial processes.
- Public Key Infrastructure.
- Class president.
- With honors.

2012 - 2017 Bachelor's degree in Industrial Electronics and Automation Engineering,Gijón Polytechnic School of Engineering:
- Specialized in Industrial Robotics.
- MENTOR Program. Providing help to new Erasmus students in the campus.
- Member of the self-evaluation committee of the Industrial Electronics and Automation Engineering degree.
- Tandem language exchange program of the University of Oviedo.
- With honors.

Publications

Substation Bill of Materials: A Novel Approach to Managing Supply Chain Cyber-risks on IEC 61850 Digital Substations
International Conference on Electrical, Computer and Energy Technologies (ICECET), January 2026

Gotta Catch 'em All: Aggregating CVSS Scores
Proceedings of the XVII Reunión española sobre criptología y seguridad de la información (RECSI), October 2022

A Novel Model for Vulnerability Analysis through Enhanced Directed Graphs and Quantitative Metrics
MDPI Sensors, March 2022

How to Quantify the Security Level of Embedded Systems? A Taxonomy of Security Metrics
Proceedings of the 2020 18th IEEE International Conference on Industrial Informatics (INDIN), July 2020, Brunsgaard A/S

Training

Embedded Systems Security (ESS)

Attacks and Countermeasures for embedded systems
December 2025

Attacks | Countermeasures | Risk Assessment | Communication Protocols | Cryptography

Workshop for cybersecurity incident response in the industry

Key strategies and tools to manage incidents in the industry
October 2025

Tabletop exercise | Attack Vectors | Incident Types | Methodologies | Remediations | Incident Notification

Cybersecurity in the design, operation and industrial maintenance in the electric industry

Protecting the Grid from cyber risks
October 2024
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OT | Vulnerabilities in the Industry | Cybersecurity from Design | Cybersecurity Requirements | Cybersecurity Risk Assessment | Metrics

TEEPwn - Trusted Execution Environments

Breaking TEEs by Experience
November 2023

Offensive | Vulnerability Identification | TEE Exploitation | ARMv8 TrustZone | Trusted Application Exploitation | Privilege Escalation

Languages

Spanish	●●●●●
English	●●●●●
Basque	●●●●○
Portuguese	●●●○○
Serbo-Croatian	●●○○○

IT skills

Linux	●●●●●
Python	●●●●○
C / C++	●●●●○
AWS Cloud	●●●●○
Assembly	●●●○○
LaTeX	●●●●○

Interests

3D Printing and design:	Filament and resin printer, miniature painting.
Electronic projects:	Arduino, PCB design, Raspberry Pi, Nixie Clock, amplifiers.
Sports:	CrossFit, Bike, hiking and Running.
Music	Bass and Guitar.
Board games	Warhammer, Yu-Gi-Oh, Catan.
Language learning	Japanese, Serbo-Croatian.